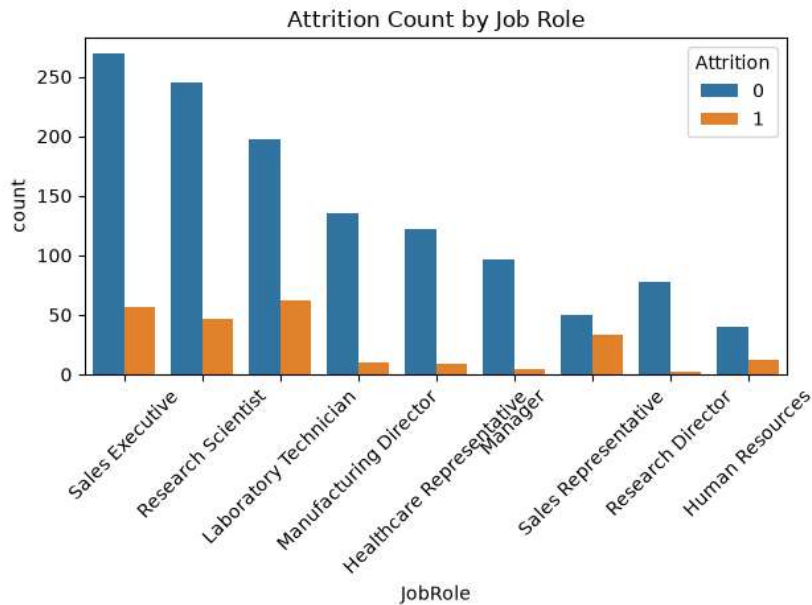


HR Employee Attrition Analysis

Full results write-up -- IBM HR Analytics dataset, 1,470 employees, 35 features

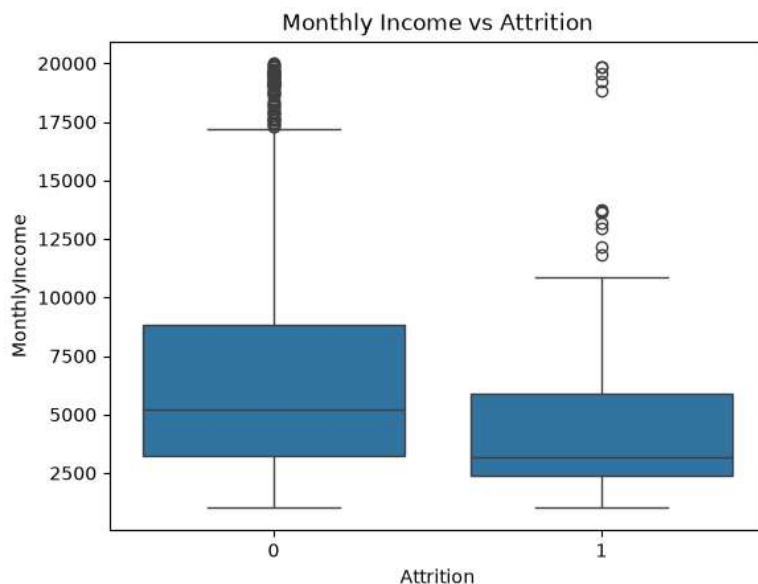
This document walks through the analysis in order: who leaves, what relates to leaving, and how well two models can predict it. Every chart here is generated directly from the dataset.

1. Who leaves? Attrition by job role



Sales Executives show the most departures in raw count, but that's mostly because it's the largest role in the dataset. Looking at rate instead of count tells a different story: Sales Representatives lose almost 40% of their people, and Laboratory Technicians lose about 24% -- both far above the 16% company-wide average, and above Sales Executives' own 17%.

2. Does pay level relate to leaving?

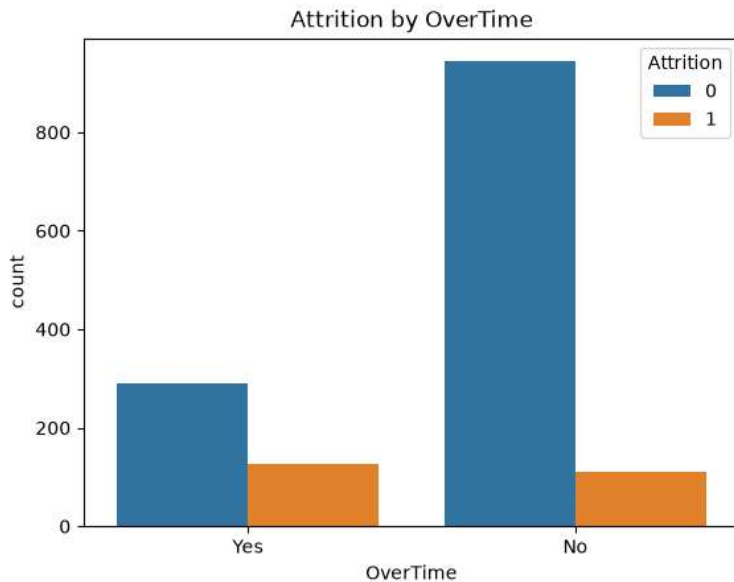


Yes, but not dramatically. Median monthly income for employees who left is Rs 3,202, against Rs 5,204 for those who stayed. The spread for leavers is also tighter -- fewer high earners walk away. It's a real signal, but a smaller one than the model ends up giving it credit for.

HR Employee Attrition Analysis

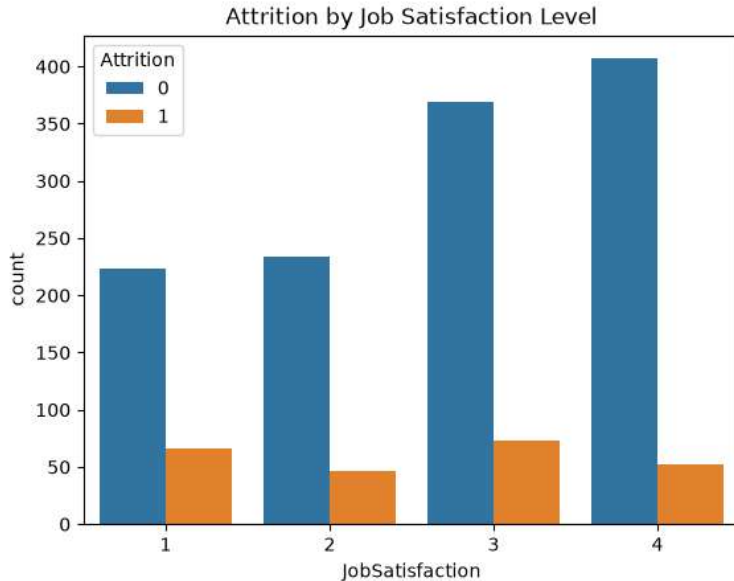
Full results write-up (continued)

3. Does overtime relate to leaving?



This is the strongest relationship in the whole dataset. Employees working overtime leave at about 31%, versus roughly 10% for everyone else -- close to 3x higher. It shows up as the top feature in both the correlation heatmap and the Random Forest importances later on, not just one method agreeing with itself.

4. Does satisfaction level relate to leaving?

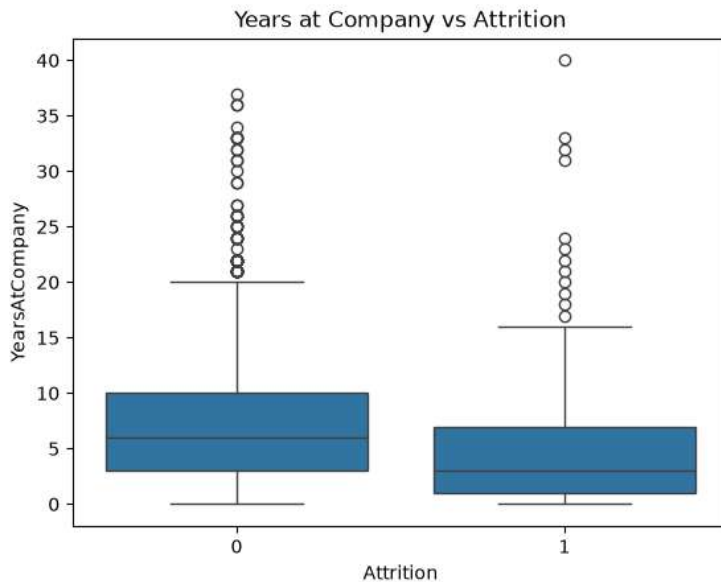


Weakly, and not in a clean straight line -- satisfaction level 3 doesn't clearly beat level 1 or 2 in the raw counts. Job satisfaction alone doesn't separate leavers from stayers as cleanly as overtime or income do; it's a softer signal on its own.

HR Employee Attrition Analysis

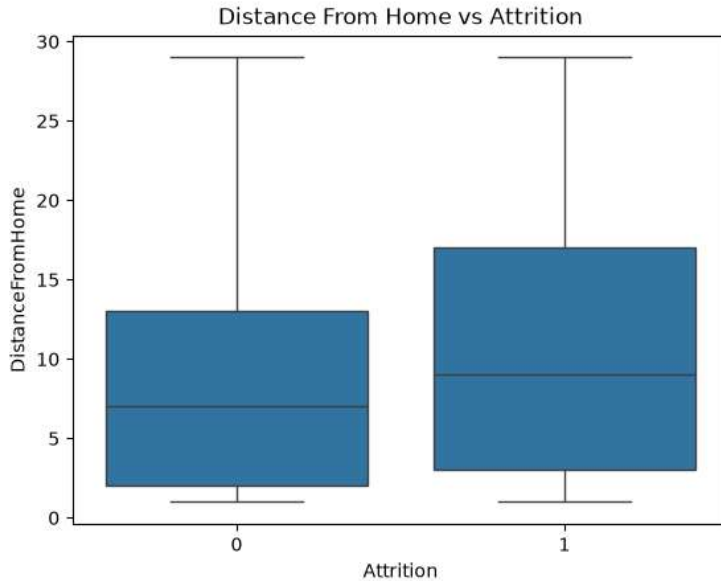
Full results write-up (continued)

5. Are leavers mostly newer employees?



Yes. Median tenure for leavers is 3 years, versus 6 years for those who stayed. This reads as an early-career retention gap concentrated in the first few years, not a problem spread evenly across tenure levels.

6. Does commute distance relate to leaving?



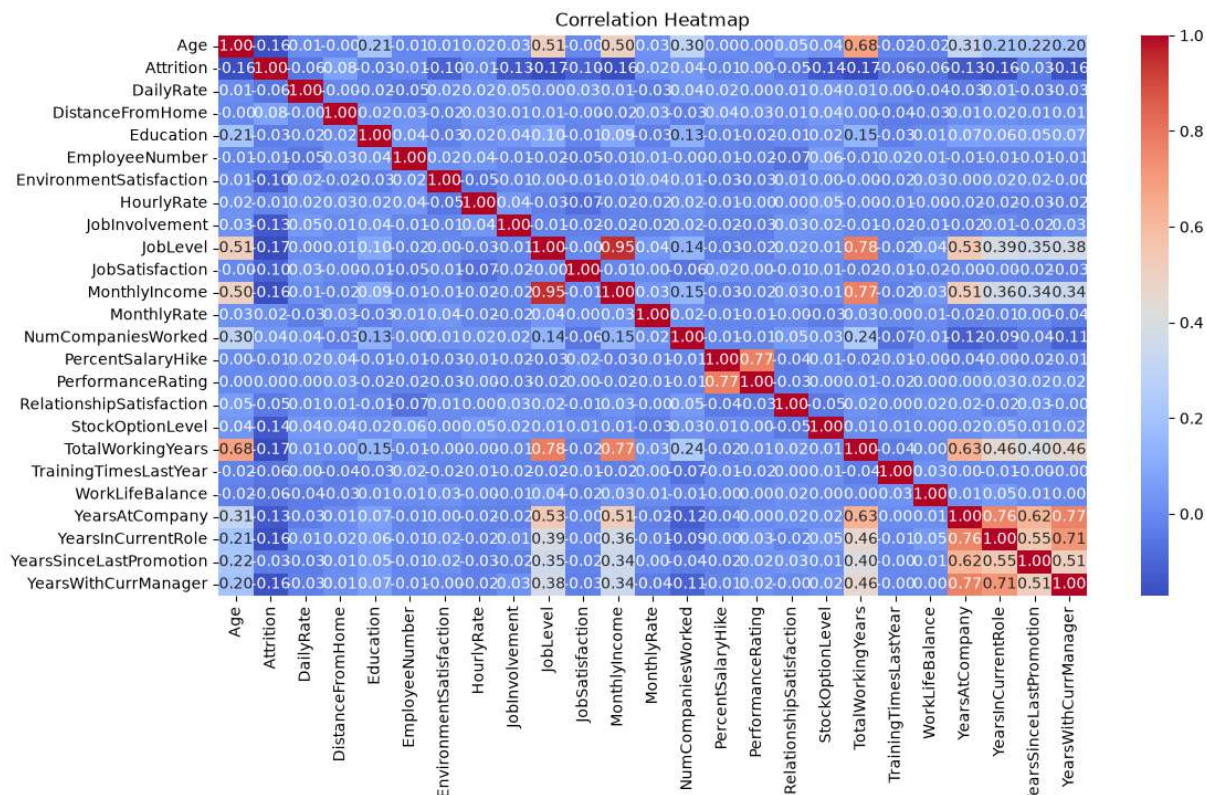
A bit. Median distance from home is 9 for leavers versus 7 for stayers, and the interquartile range is wider for leavers. It's a real but minor factor -- nowhere near the effect size of overtime or tenure.

HR Employee Attrition Analysis

Feature relationships and model-driven importance

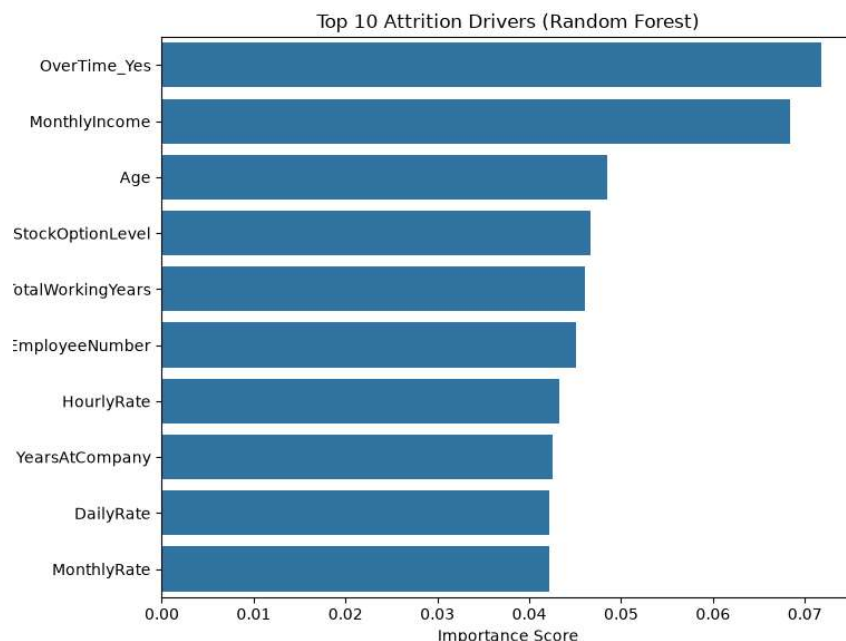
7. Correlation heatmap

A full numeric scan confirms the same story: OverTime, MonthlyIncome, and tenure-related fields (YearsAtCompany, YearsInCurrentRole, YearsWithCurrManager) are the fields most associated with Attrition, while things like DailyRate or PerformanceRating barely move it.



8. Top attrition drivers (Random Forest)

The model agrees with the manual EDA: OverTime and MonthlyIncome sit at the top by a clear margin, ahead of Age, StockOptionLevel, and TotalWorkingYears.



HR Employee Attrition Analysis

Model comparison and recommendations

9. Logistic Regression vs. Random Forest

	Accuracy	Recall (leavers, class 1)	Precision (leavers, class 1)
Logistic Regression	72%	56%	25%
Random Forest	86%	21%	42%

Random Forest wins on accuracy but that's the wrong number to optimize here -- it only catches 21% of employees who actually leave. Logistic Regression looks weaker overall (72% accuracy) but catches 56% of leavers, more than double the rate. For a tool meant to flag at-risk employees before they quit, recall on the leaver class is what matters, and Logistic Regression is the better choice despite the lower headline score.

Recommendations

- Treat overtime as the primary lever, not a side factor -- it has the largest, most consistent effect across the correlation analysis and the model.
- Concentrate retention efforts on the 0-3 year tenure window, where the actual risk sits, rather than spreading resources company-wide.
- For an attrition-flagging tool, prioritize recall over accuracy -- Logistic Regression is the better model here despite its lower headline score.
- Within roles, look at rate rather than raw headcount -- Sales Representatives and Lab Technicians are the highest-risk roles, not Sales Executives.

What I'd want next

Exit interview text, manager notes, or engagement survey scores would probably explain more than anything in this dataset. The numeric HR fields tell you who's leaving. They're weaker at telling you why.